

3/14 CHORD IN TERMS OF GIVEN POINTS.

Find the equation of the chord of contact of tangents from (x_1, y_1) to the (i) parabola $y^2 = 4ax$

(ii) ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

IN (iii) hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

Soln- Let P be the given pt. (x_1, y_1) and let the tangent from P to the parabola, touch the parabola at Q and R, then

QR is called the chord of contact of tangents (x_1, y_1)

Let Q be (x_2, y_2) and R be (x_3, y_3)

The eq. of the tangent at Q (x_2, y_2)

$$yy_2 = 2a(x + x_2)$$

Since the tangent passes through (x_1, y_1) , we have

$$y_1 y_2 = 2a(x_1 + x_2) \quad \text{--- (1)}$$

Eq. of tangent at R (x_3, y_3)

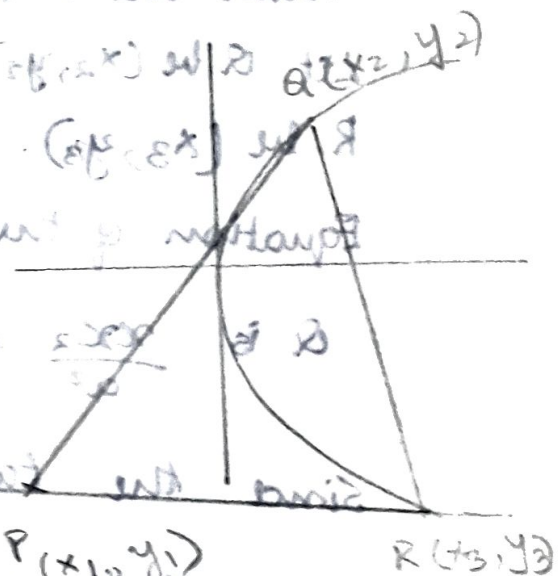
$$yy_3 = 2a(x + x_3)$$

Since the tangent passes through (x_1, y_1) , we have

$$y_1 y_3 = 2a(x_1 + x_3) \quad \text{--- (2)}$$

Consider the eq.

$$yy_1 = 2a(x + x_1) \quad \text{--- (3)}$$



Eq (3) is first degree equation in x & y and hence represent a straight line.
 By conditions (1) & (2), here the line (3) passes through the point (x_2, y_2) and (x_3, y_3) .
 Therefore eq (3) is the chord of contact of the tangents from (x_1, y_1) to the parabola.

Q(ii) Let P be the given point (x_1, y_1) .

Let the tangent from P to the given ellipse

touch the ellipse at Q and R.

Let Q be (x_2, y_2) and

R be (x_3, y_3) .

Equation of the tangent at

$$Q \text{ is } \frac{xx_2}{a^2} + \frac{yy_2}{b^2} = 1.$$

Since the tangent passes through (x_1, y_1) we have

$$\frac{x_1 x_2}{a^2} + \frac{y_1 y_2}{b^2} = 1 \quad \text{--- (1)}$$

Equation of tangent at R is

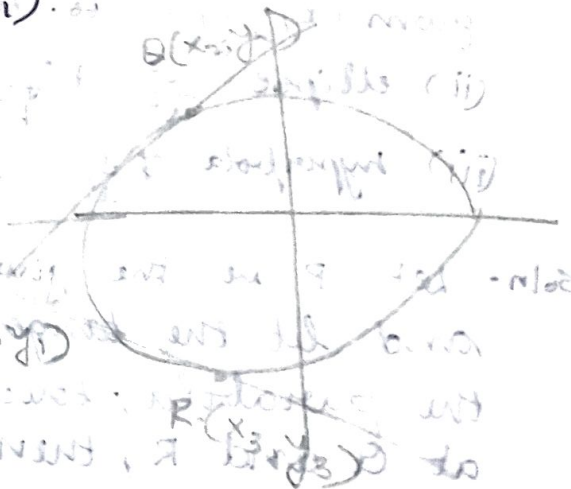
$$\frac{x x_3}{a^2} + \frac{y y_3}{b^2} = 1.$$

Since the tangent passes through (x_1, y_1) we have

$$\frac{x_1 x_3}{a^2} + \frac{y_1 y_3}{b^2} = 1 \quad \text{--- (2)}$$

Consider the equation

$$\frac{x x_1}{a^2} + \frac{y y_1}{b^2} = 1 \quad \text{--- (3)}$$



Equation (3) is a first degree equation in x, y .
 and hence represent a straight line.

By conditions (1) & (2), we say that the line (3) passes through (x_2, y_2) and (x_3, y_3) . Eqn

$\therefore$ Eq. (3) represent the chord of contact from of tangents from (x_1, y_1) to the ellipse.

Q2. Find the chord coordinates of the point of intersection of tangents drawn to the parabola $y^2 = 4ax$ at the points where it is cut by the straight line $x \cos \alpha + y \sin \alpha = p$.

Ans- Let the given line cut the parabola at P and Q. Then the equation to PQ is $x \cos \alpha + y \sin \alpha = p$

— (1)

Let the tangent at P and Q intersect at (x_1, y_1) . Then PQ is the chord of contact of tangents from (x_1, y_1) .

Eq. to PQ is $yy_1 = 2a(x+x_1)$

$$yy_1 = 2ax + 2ax_1$$

$$2ax + yy_1 = 2ax_1 \quad \text{--- (2)}$$

(1) & (2) are identical.

$$\frac{-2a}{\cos \alpha} = \frac{y_1}{\sin \alpha} = \frac{2ax_1}{p}$$

$$y_1 = -2a \tan \alpha$$

$$x_1 = -p \sec \alpha$$

$\therefore$ The pt. (x_1, y_1) is $(-p \sec \alpha, -2a \tan \alpha)$.

Q3.

P.T. the tangent we draw to the parabola $y^2 = 4ax$ from any pt. on the straight line $x + 4a = 0$, the chord of contact subtend a right angle at the vertex of the parabola.

Soln:

Given parabola is

$$y^2 = 4ax \quad \text{--- (1)}$$

Any point on the line

$$x + 4a = 0 \text{ is}$$

$$(-4a, y_1).$$

Eq. of the chord of contact of tangents from

(x_1, y_1) to the parabola is

$$yy_1 = 2a(x + x_1)$$

Eq. of the chord of contact of tangents from $(-4a, y_1)$ to the parabola is

$$yy_1 = 2a(x - 4a)$$

$$yy_1 = 2ax - 8a^2$$

$$2ax - yy_1 = 8a^2$$

$$\frac{2ax - yy_1}{8a^2} = 1 \quad \text{--- (2)}$$

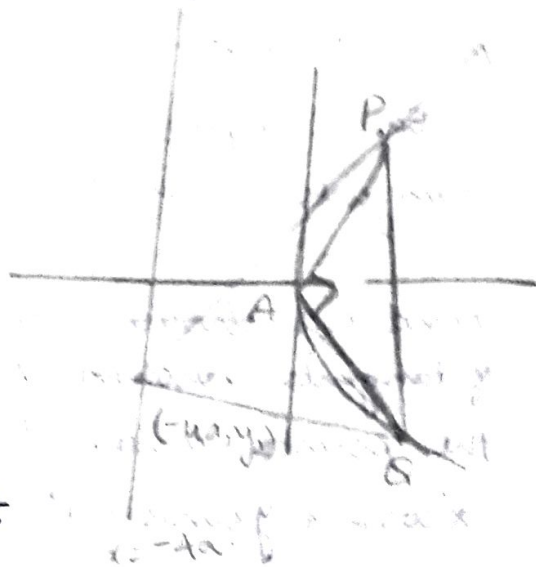
The combined eq. of AP & BP is obtained by homogenising (1) using (2).

$$y^2 = 4ax$$

$$y^2 = 4ax \times 1$$

$$y^2 = 4ax \times \frac{2ax - yy_1}{8a^2} = 1$$

$$2ay^2 = 2ax^2 - xyy_1 \quad (\text{if } x = 0 \text{ then } y = 0)$$



$$\text{ie } 2ax^2 - y_1xy - 2ay^2 = 0.$$

$$\text{coeff of } x^2 + \text{coeff of } y^2: 2a - 2a = 0.$$

$\therefore AP$ and AQ are perpendicular.

$$\therefore \angle PAQ = 90^\circ.$$

(84) The chord PA of an ellipse subtend a right angle at the centre of the ellipse. S.T the locus of the point of intersection of tangents at P and A is

$$\text{the ellipse } \frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{1}{a^2} + \frac{1}{b^2}$$

Soln Let the given ellipse be

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \text{--- (1)}$$

Let the tangent at P and A intersect at (x_1, y_1) then

PA is the chord of contact of tangents from (x_1, y_1) .

Therefore the eq. to PA is

$$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1 \quad \text{--- (2)}$$

Therefore the combined eq. of CP & CA is obtained by homogenising (1) by (2).

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

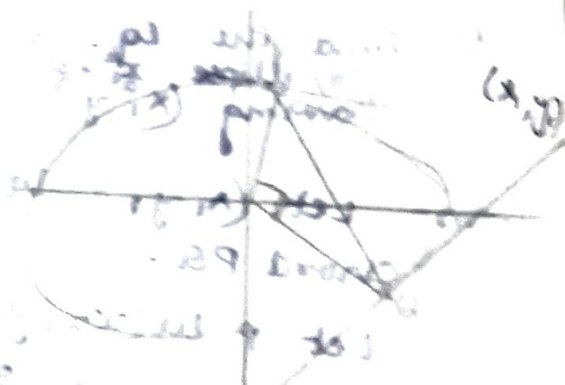
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \left(\frac{xx_1}{a^2} + \frac{yy_1}{b^2} \right)^2 \frac{a^2 b^2}{a^2 b^2}$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{x^2 x_1^2}{a^4} + \frac{2xx_1yy_1}{a^2 b^2} + \frac{y^2 y_1^2}{b^4}$$

$$\left(\frac{1}{a^2} - \frac{x_1^2}{a^4} \right) x^2 - \frac{2x_1 y_1}{a^2 b^2} xy + \left(\frac{1}{b^2} - \frac{y_1^2}{b^4} \right) y^2 = 0 \quad \text{--- (3)}$$

which is the combined eq. of CP and CA .



Since $\angle PCQ = 90^\circ$, CP and CQ are perpendicular

$\therefore$ In eq (3) Coeff of x^2 + Coeff of $y^2 = 0$.

$$\frac{1}{a^2} - \frac{x_1^2}{a^4} + \frac{1}{b^2} - \frac{y_1^2}{b^4} = 0$$

$$\frac{x_1^2}{a^4} + \frac{y_1^2}{b^4} = \frac{1}{a^2} + \frac{1}{b^2}$$

Therefore the locus of (x_1, y_1) is

$$\frac{x^2}{a^4} + \frac{y^2}{b^4} = \frac{1}{a^2} + \frac{1}{b^2}$$